

Ethical Research Writing

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Ethical & Professional Vocabulary

- Values, Morals, Ethics, Principles, Professional Ethics, Code of Ethics, Integrity, Honesty, Accountability, Responsibility, Norms, Law, Compliance, Etiquette

- What would you do?
  - You obtain a result that supports your hypothesis
  - Later, you discover that one experimental setting was incorrect
  - Nobody else knows about the error
  - Correcting it will delay the paper
  - The submission deadline is soon
  - Your supervisor expects the paper to be submitted

- In order to make a decision, we need to understand all the vocabulary.



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Values: What is considered important?

- Values are the qualities, goals or outcomes that is considered important
  - Safety, Success, Knowledge, Fairness, Innovation, Independence, Sustainability, Social impact

- **Individual**
  - What do I consider important?
  - An engineer may value growth and mastery

- **Organisational**
  - What does the organisation consider important?
  - A company may value rapid innovation, customer safety and profitability

- Values influence priorities and decisions
  - A researcher who values **accuracy** repeats and validate an experiment



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Morals: What is believed to be right or wrong?

- Morals are beliefs about what is right, wrong, good or unacceptable
  - Influenced by: Family, Culture, Religion, Education, Society, Personal experience
  - Create Moral disagreement
- **Individual**
  - What do I believe is right or wrong?
- **Collectiveness**
  - What does a community or organisation consider acceptable or unacceptable?
- Moral beliefs originate in people
  - Organisations develop a collective moral culture through the beliefs and decisions of their leaders and members
- Good intentions alone may not resolve moral disagreement
  - Ethical reasoning is needed



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Ethics: How should decisions be made?

- Systematic examination of how a person or organisation should decide and act
  - Ethical reasoning considers
    - Benefits, Rights, Fairness, Duties, Consequences, Professional standards
- **Individual**
  - How should I decide and act?
  - A student must decide whether to disclose an experimental error
- **Organisational**
  - How should the organisation make and justify its decisions?
  - A university must decide how allegations of research misconduct should be investigated fairly



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Ethics: How should decisions be made?

- Personal preference:
  - I do not like this decision
- Moral belief:
  - I believe this decision is wrong
- Ethical justification:
  - This decision is wrong because it hides risks from people, doesn't get their clear consent, and treats them unfairly
- An ethical conclusion should be supported by reasons rather than preference or belief



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Principles: What fundamental rules guide conduct?

- Principles are fundamental rules or standards used to guide decisions and behaviour
  - Honesty, Fairness, Transparency, Confidentiality, Reproducibility
- **Individual**
  - What principles guide my behaviour?
  - A researcher follows the principle of transparency by reporting all important assumptions
- **Organisational**
  - What principles guide the organisation's policies and decisions?
  - A journal follows the principle of fairness by evaluating submissions using consistent criteria
- Principles convert values into action
  - Value: Accuracy is important
  - Principle: Results must be verifiable
  - Action: Repeat uncertain experiments



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Professional ethics: What conduct does a profession expect?

- Principles are fundamental rules or standards used to guide decisions and behaviour
  - Honesty, Fairness, Transparency, Confidentiality, Reproducibility
- **Individual**
  - What conduct is expected from me as a researcher, engineer, teacher or reviewer?
- **Organisational**
  - What standards should the profession and its institutions maintain?
- Examples: Report truthful evidence, Maintain confidentiality, Disclose conflicts of interest, Give appropriate credit, Correct significant errors
- Professional roles create additional obligations
  - A reviewer should protects the confidentiality of an unpublished manuscript
  - A government assigns national priority work only to competent gov agencies



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Code of ethics: What written standards govern conduct?

- A code of ethics is a formally documented set of ethical standards
  - Adopted by a profession, institution or organisation
- **Individual**
  - What written standards am I expected to follow?
- **Organisational**
  - What standards has the organisation adopted, communicated and enforced?
- Examples: IEEE Code of Ethics, University research-integrity policy, Journal publication-ethics policy, Company code of conduct
- Code of ethics provides common expectations



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Honesty: Is information presented truthfully?

- Honesty means representing information truthful and accurate
- **Individual**
  - Am I presenting information truthfully?
  - An author reports that the proposed method required validation for certain scenarios
- **Organisational**
  - Is the organisation communicating accurately with employees, customers, and the public
  - A company does not claim that a vehicle is “completely safe” based only on simulation results
- Honesty includes avoiding misleading omissions
  - A statement can be technically true but still misleading because of omitted information
    - E.g. A paper reports a 20% improvement but does not disclose that the comparison used a deliberately weak baseline



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Responsibility: Who has the duty to act?

- Responsibility is the duty or obligation to perform a task, role or take appropriate action
- **Individual**
  - What duties am I expected to perform?
  - A supervisor provides appropriate guidance
- **Organisational**
  - What duties must the organisation assign, support and fulfil?
  - A laboratory provides safe equipment, training, data storage and clear research procedures
- Responsibility must be clearly assigned



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Accountability: Who must answer for the outcome?

- Accountability is the obligation to justify and answer for decisions, actions and outcomes
- **Individual**
  - Can I explain my decision and accept its consequences?
- **Organisational**
  - Can the organisation identify who made the decision, examine what happened and take corrective action?
- Accountability requires more than accepting blame
  - The organisation maintains systems that allow decisions to be examined and corrected

Responsibility	**Accountability**
Can be distributed among people	Must remain identifiable
“Who should do it?”	“Who must explain what happened?”



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Norms: What behaviour is commonly expected?

- Norms are informal expectations about behaviour within group, community, organisation
- **Individual**
  - What behaviour is expected from a person in this group?
- **Organisational**
  - What behaviour does the organisation encourage, tolerate or discourage through its culture?
- **Examples:** Citing previous work, Responding to important communication, Arriving on time, Acknowledging collaborators, Allowing others to speak

- Common behaviour is not automatically ethical and not followed blindly



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Law: What is legally required or prohibited?

- Law consists of rules created and enforced by recognised legal authorities
- **Individual**
  - What am I legally required or prohibited from doing?
- **Organisational**
  - What legal obligations must the organisation satisfy?
- **Examples:** Copyright, Data protection, Employment law, Anti-discrimination law

- Legal conduct is not always ethical conduct
  - Law establishes minimum enforceable requirements



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Others

- Compliance: Are applicable requirements being followed?
  - Compliance is the process of following applicable laws, regulations, standards, policies and procedures

- Etiquette: What behaviour is respectful and appropriate?
  - Etiquette refers to socially or professionally appropriate ways of interacting with others



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Terms

Term	General question
Values	What is considered important?
Morals	What is believed to be right or wrong?
Ethics	How should decisions be made and justified?
Principles	What fundamental rules guide conduct?
Professional ethics	What conduct does a profession expect?
Code of ethics	What written ethical standards apply?
Integrity	Do actions match stated values?
Honesty	Is information presented truthfully?
Responsibility	Who has the duty to act?
Accountability	Who must answer for the outcome?
Norms	What behaviour is commonly expected?
Law	What is legally required or prohibited?
Compliance	Are applicable requirements being followed?
Etiquette	What behaviour is respectful and appropriate?



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How to Write a Professional Email



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Cold emails to professors or collaborators

- Dos:
  - Start with context: Mention how you found their profile or paper
    - “I recently read your paper on cooperative perception in IEEE TVT and found your work very relevant to my interests in vehicular networks”
  - Explain what you offer:
    - “I have experience implementing DCC algorithms in ns-3 and would like to explore integrating them with your ongoing work on V2X perception”
  - Show complementarity:
    - “I believe my background in communication-layer modeling could contribute to your work on semantic-aware V2X systems”
  - Close politely and concisely

- Don’ts:
  - Never start with “I’m looking for an opportunity” or “Please give me a project ”
  - Avoid generic emails: always personalize with the recipient’s research



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Reply Emails

- Rule 1: Answer the questions directly first
- Rule 2: Then ask your follow-up queries clearly
- Rule 3: Be polite and structured

- Follow a three-step rule:
  - **Answer** → “Yes, I’ll send the report by Friday.
  - **Clarify** → “Would you prefer the updated figures as well?”
  - **Appreciate** → “Thank you for your guidance.

- **Politeness markers:**
  - “I appreciate your time.
  - “Please let me know if you have further suggestions”



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How to Attend and Contribute in Meetings



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Meetings

- If you're an attendee: listen silently, avoid interrupting
- If you need to ask something: Wait for the speaker to finish
- Say, "Apologies for the interruption, I have a clarification on slide 5"
- While asking questions: First appreciate the talk:
  - "Thank you for the insightful presentation…"
- Then ask for specific clarifications
- Keep a notebook or annotate slides note down slide numbers for discussion
- During discussion: Avoid acknowledging by saying "hmm," "okay"



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How to understand research papers

How to select the right scope

How to identify a genuine gap

How much literature is enough

What to do after selecting a topic



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Choosing a Research Topic

- Domain → Application → Problem → Context → Question → Contribution → Evaluation

Stage	Communication Example	Mechanical / Battery Research Example
Domain	Communication Systems	Battery Tech & Thermal Management
Application	V2X (Vehicle-to-Everything)	Electric Vehicle Fast-Charging Packs
Problem	Network Congestion	Capacity Fade & Thermal Runaway
Context	Dense Traffic Environments	Extreme Ambient Temp + High C-Rate (4C)
Question	Which information should be selected?	How to optimize cold-plate coolant channel geometry?
Contribution	New Selection Mechanism	Novel Micro-channel Cold Plate with Hybrid PCM
Evaluation	Channel Load & Situation Awareness	Cell Temperature Uniformity, Pressure Drop & Cycle Life

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Refining Topic Scope: Field vs. Task

- Broad Topics Name a Field; Researchable Topics Name a Task

Scope Level	Example Topic	Methodological Analysis
TOO BROAD	Artificial intelligence for automated vehicles	“Artificial intelligence for automated vehicles” could include perception, prediction, planning, control, human interaction and communication.
STILL BROAD	Object detection for automated vehicles	Focus on a subfield, still lacks specific operation
RESEARCHABLE	Improving pedestrian detection under low illumination	Identifies a specific problem, concrete operating context (low light), and clear, measurable outcomes.
TOO NARROW	Changing one threshold for one video without broader justification	Overly isolated parameter tweak; fails to generate transferable or generalizable academic knowledge.

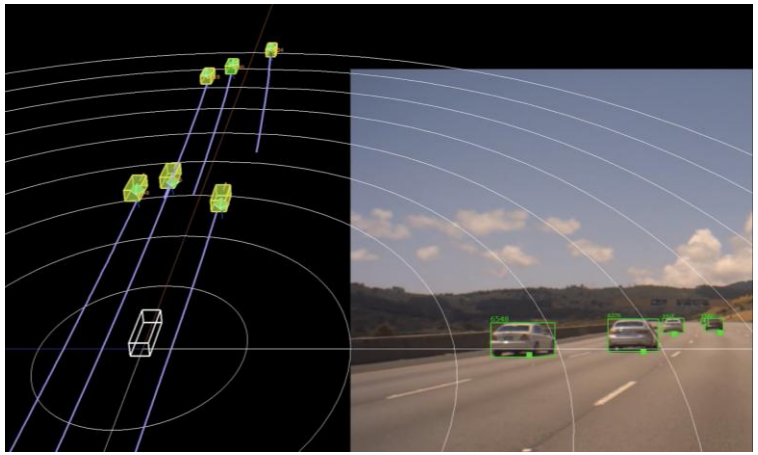
- **A good topic should be**
  - **narrow enough to investigate rigorously**
  - **but broad enough beyond one artificial case**



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How Research Areas Branch: The Camera Story

- **Camera Hardware**
  - Improving the physical process of capturing accurate photographs
- **Digital Image Formation**
  - Digital sensors, signal processing & computational photography
- **Computer Vision**
  - Extracting meaningful information and features from pixel data
- **Object Detection**
  - Classification, bounding boxes, segmentation, and tracking
- **Automated-Vehicle Perception**
  - Real-time environmental sensing for safe driving decisions



- **Occlusion & Limited Sensing**
  - Overcoming physical line-of-sight limits (e.g., trucks blocking pedestrians)
- **Cooperative Perception & V2X**
  - Connecting vehicles and infrastructure to extend sensing beyond line-of-sight
- **Key Insight:** New fields rarely appear independently, they emerge when established technologies encounter new requirements or physical limitations

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Decide the Level at Which You Want to Contribute

Contribution Level	Operational Focus	Primary Examples
Component/ theoriotical	Isolated algorithm, link-layer, or perception performance	Object detection, channel estimation, coding schemes, standalone trajectory prediction
System	Interactions across vehicles, protocol mechanics, and resource distribution	Resource scheduling, congestion control, V2X information exchange protocols
Application / End-to-End	Closed-loop vehicle behavior, cooperative control, and safety outcomes	Cooperative merging, automated collision avoidance, platooning decisions



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Choosing a Research Topic

Factor	Assessment Focus	Red Flag / Failure Mode
1. Research Significance	Addresses a meaningful research gap beyond superficial novelty.	High novelty, but zero theoretical or practical utility.
2. Student Skills & Interest	Aligns with current strengths or reasonable learning curves.	Core work requires massive skill acquisition outside project timelines.
3. Lab & Supervisor Expertise	Leverages local domain knowledge, mentorship, and prior work.	Complete isolation with no internal guidance or subject matter experts.
4. Data, Tools & Resources	Hardware, datasets, software licenses, and sensors are currently accessible.	Unobtainable datasets, proprietary access barriers, or unfunded gear.
5. Time & Evaluation Feasibility	Scoped tightly enough to gather data, evaluate, and write up before deadline.	Requires long-term longitudinal studies or open-ended mathematical proofs.



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Research Topic Novelty vs. Saturation

- A Mature Field Is Not Necessarily an Exhausted Field
- **Contextual Shifts:** A problem solved in static or centralized setups reopens entirely under high mobility, decentralized control, or dynamic topologies
- **Scale & Density:** Algorithms optimized for sparse or small-scale settings often break down non-linearly when subjected to ultra-dense deployment
- **Integrated Requirements:** Emerging applications merge previously separate domains such as tightly coupling high-bandwidth sensing perception with strict real-time communication bounds
- **Real-World Bottlenecks:** Shifting focus from pure algorithmic accuracy to system-level constraints, including compute latency, bandwidth efficiency, energy, and security



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Research Topic Novelty vs. Saturation

- A Mature Field Is Not Necessarily an Exhausted Field
  - A Gap Is an Unresolved Reason for a New Study
  - “Future Work” Is a Clue, Not a Gap Certificate

- **Contextual Shifts:** A problem solved in static or centralized setups reopens entirely under high mobility, decentralized control, or dynamic topologies
- **Integrated Requirements:** Emerging applications merge previously separate domains such as tightly coupling high-bandwidth sensing perception with strict real-time communication bounds

- Try to understand
  - Domain → Application → Problem → Context → Question → Contribution → Evaluation



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Understanding types of Research Papers



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Types of Academic Publications

- **Original Research (Full Paper)**
  - Presents a complete, validated novel contribution and methodology
- **Survey / Review Paper**
  - Synthesizes existing literature, establishes a taxonomy, and outlines open challenges
- **Tutorial**
  - Teaches established or new technology
- **Poster / Short / Work-in-Progress Paper**
  - Shares early-stage ideas, preliminary results, or initial proof-of-concepts
- **Benchmark & Dataset Paper**
  - Provides reusable datasets, standardized comparisons to support the community



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Publication Routes

Route	General role
Journal/Transactions	Full-length, peer-reviewed contribution
Letters	Short and focused contribution
Conference proceedings	Timely peer-reviewed work; importance varies by field
Magazine	Accessible overview, application or tutorial content
Preprint archive (arXiv)	Rapid public dissemination; not automatically peer reviewed



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Publication Routes

Parameter	General Journal	Transactions
Primary Focus	Broad discipline, interdisciplinary research, or emerging cross-domain trends.	Highly specialized, deep technical contributions within a specific sub-field.
Target Audience	Broad scientific community, cross-field researchers, and general practitioners.	Domain specialists, specific technical society members, and niche research groups.
Scope & Breadth	High breadth; covers multiple sub-fields or broad topics (e.g., IEEE Access, Nature, IEEE IoT Journal).	High depth; tightly scoped to a specific technical domain (e.g., IEEE Transactions on Vehicular Technology).
Review Emphasis	Broad relevance, overall impact, cross-disciplinary utility, and readability.	Mathematical/methodological rigor, domain depth, and technical correctness.
Publication Model	Often uses open-access or rapid-publication formats appealing to wide audiences.	Traditional multi-round archival review focused on establishing permanent domain standards.



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- Q1 and Q2 Rank Journals
- IEEE, ACM, Elsevier
- Socities
- Standards: ETSI, IEEE, SAE



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How to write CV or Resume



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CV, Resume and Biodata

- Resume
  - Short, selective and tailored
  - Common in US and Canada for industry applications
    - Usually one or two pages
- Curriculum vitae
  - Comprehensive record of academic work
  - Publications, teaching, grants, conferences and service
    - Length grows with the person's career
- Biodata
  - Personal and demographic information
  - Not a substitute for a professional resume
    - specific regional/government contexts
- Always follow the terminology used in the advertisement



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Industry and Academia

- Industry resume:
  - What problem can this candidate solve?
  - What skills can the candidate apply immediately?
  - What measurable value has the candidate created?
- Academic CV:
  - What knowledge has this candidate produced?
  - What and how has the candidate taught?
  - Can the candidate develop an independent research programme?
  - How will the candidate contribute to the department?
- Same experience, different emphasis
  - Company: result, scale, efficiency and business relevance
  - Academia: method, originality, publication and scholarly contribution



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Industry and Academia

- PhD student developed a machine-learning model for crop disease detection

- Industry resume emphasise on:
  - model performance, deployment, product delivery, and cross-functional teamwork
- Academic CV emphasise on:
  - publications, research methods, datasets and contribution to agricultural science

- Neither version is dishonest
  - Highlights different facets for the same authentic technical effort



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Recruitment

- Industry Recruitment
  - **Screening:** eligibility checks, filtering or prioritization.
  - **Review & Assessment:** Recruiter/hiring manager review and technical skills screening
  - **Interviews & Offer:** Candidate interviews and final job offer
- Academic Recruitment
  - **Dossier & Portfolio Review:** Academic CV, publication record, and research statements
  - **Peer Shortlisting:** Reference letters and departmental committee meeting
  - **Presentation & Approval:** Seminar or teaching demo and committee recommendation



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Industry Filtering Mechanism

- Applicant Tracking System (ATS)
  - Primarily a recruitment platform
  - Organise candidate information and support the hiring workflow
  - Different employers enable different functions
- ATS Role
  - Publish job advertisements, Store candidate profiles, Search and filter candidates
  - Move candidates through recruitment stages, Schedule interviews and record feedback
- ATS Additional Roles
  - Match scores, Candidate recommendations, AI-generated summaries
- There is no single universal ATS algorithm



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ATS Resume Parsing

- Text extraction
  - DOCX, PDF, Scanned PDF
- Parsing
  - Converts unstructured text into structured data
  - Standard patterns (email, phone numbers) are easy to extract via rules
  - Contextual data (skills, titles) requires complex NLP
- Normalisation
  - “ML” → machine learning, “SDE” → software development engineer
  - Use Dictionaries and Machine-learning models
- Good Practise:
  - Include both the full term and standard acronym (e.g., "Machine Learning (ML)")
  - Avoid non-standard abbreviations
  - Contextualize skills within bullet points rather than relying solely on a standalone list
    - Listing “Python” once in a skill list is weaker than explaining where and how Python was used



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Knockout Questions

- Knockout questions are non-negotiable conditions
- Common examples
  - Are you legally authorised to work in this country?
  - Are you willing to work at the stated location?
  - Do you have a minimum number of years of experience?
- India-specific examples may include:
  - Notice period, Current and expected CTC, Willingness to relocate, etc
- An unsatisfactory answer may:
  - Auto-reject the application
  - Add a tag
  - Move it to a separate queue
  - Require manual review



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Keyword-Relevance Algorithm

- Search engines may consider:
  - Exact term match, Term frequency, Phrase match, Proximity between related words
- Location of the term may also important
  - “Python” in recent work experience can be more meaningful
    - than “Python” in a long list of unverified skills
- More repetitions do not always produce a better result
  - Some search tools consider
    - how often a term occurs in the document
    - how informative it is across the collection
    - whether the document is unusually long
- Useful context is normally better than uncontrolled repetition
  - Do not add more terms, present the “known” terms in a better way



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Strong Candidate May NOT be recruited

- The application arrived after shortlisting had started
- Ghost jobs
- The recruiter reviewed only the first manageable group
- The document did not parse correctly
- Relevant evidence used unfamiliar terminology
- Internal or referred candidates were reviewed first
- The vacancy was paused, changed or filled
- The profile matched another role better
- The recruiter searched only the most recent experience



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ATS Myths

- Myth: A high “ATS score” guarantees an interview
  - Reality: eligibility, competition and human judgement still matter
- Myth: PDF files are always rejected
  - Reality: readable PDFs are widely supported; follow the portal instructions
- Myth: More keywords always improve ranking
  - Reality: relevance and evidence matter more than repetition
- Myth: Graphics make a resume stand out to the ATS
  - Reality: important text inside graphics may not be extracted
- Myth: The ATS makes the final hiring decision
  - Reality: it usually supports a larger human workflow



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Industry and Academia

- Industry resume:
  - What problem can this candidate solve?
  - What skills can the candidate apply immediately?
  - What measurable value has the candidate created?

Both ATS and Human should understand the resume

- Academic CV:
  - What knowledge has this candidate produced?
  - What and how has the candidate taught?
  - Can the candidate develop an independent research programme?
  - How will the candidate contribute to the department?
- Same experience, different emphasis
  - Company: result, scale, efficiency and business relevance
  - Academia: method, originality, publication and scholarly contribution



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Some points

- First: Experience or academics
- Professional Summary / Career Objective
- India vs Abroad
- Additional necessary documents
- Single column or stylist resume
- File names (name, role)
- Consistent terms
- Text box, tables, images
- Photo
- LaTex or Word



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Latex

- LaTeX is a document preparation system based on the TeX typesetting program
- Write plain text code with markup tags with specify structure and formatting
  - Compiler turns it into a professionally formatted PDF
- Content vs. Style: Focus on writing text and equations
  - LaTeX handles page layout, margins, and font scaling automatically.
- Commands and Environments: Formatting is applied using backslash commands
  - (e.g., \section{Introduction}) and block environments (e.g., \begin{document}).
- Math Mode: Mathematical symbols and equations are rendered cleanly
- **Get Started:**
  - **Online Editing:** Beginners can use Overleaf, a cloud-based editor
  - **Local Installation:** Advanced users install softwares TeX Live or MiKTeX
- **More Information:**
  - *https://www.overleaf.com/learn/latex/Learn_LaTeX_in_30_minutes*



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Thank you for your attention

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